

1.1.5 Suppose that $\mathbb{Z}/n\mathbb{Z}$ is, let e be the identity, denote e as $e' + n\mathbb{Z}$ ($e' \in \mathbb{Z}$).
 thus $(1 + n\mathbb{Z}) \times (e' + n\mathbb{Z}) = 1 + n\mathbb{Z}$ implies $e' \equiv 1 \pmod{n}$
 let $b + n\mathbb{Z}$ be the inverse of $a + n\mathbb{Z}$, thus $0b \equiv e' \equiv 1 \pmod{n}$, Contradict.
 thus $\mathbb{Z}/n\mathbb{Z}$ isn't a multi. group. 70+10

1.1.7 Firstly, $x * y = \{x+y\} \in [0,1) \subseteq G$, thus operation $*$ is conserved by G
 $(x * y) * z = x + y - [x+y] + z = [x+y - [x+y] + z] = x + y + z - [x+y+z]$
 $= x + y + z - [y+z] + [x+y+z - [y+z]] = x * (y * z)$
 thus $*$ is associative.

$(x * y) = x + y - [x+y] = y * x$, thus $*$ is commutative.
 Secondly, the identity is 0. because $x * 0 = x + 0 - [x] = x$ ($0 * x = x * 0 = x$, definitely)
 for $\forall x$, its inverse is $\begin{cases} 1-x, & x > 0 \\ 0, & x = 0 \end{cases}$, obviously.

To sum up, G is a well-defined Abelian group in terms of $*$. ✓

1.1.20. Suppose that the order of x is n , that is, n is the smallest i s.t. $x^i = e$.

thus, ~~$(x^{-1})^n = x^{-n}$ (due to $(f^{-1})^n = (f^n)^{-1}$)~~

$$(x^{-1})^n = (x^{-1})^n x^n = x^{-1} * x^{-1} * \dots * x^{-1} * x * x * \dots * x = e$$

Also, suppose that $(x^{-1})^t = e$, by the proposition that $(x^{-1})^{-1} = x$
 and the same reasoning as the previous one, we have $x^t = e$, thus $t \geq n$.
 Sequentially, the order of x^{-1} is also n . |x| = \infty ?

1.1.29 $\nRightarrow A \times B$ is Abelian $\Leftrightarrow \forall (a_1, b_1), (a_2, b_2) \in A \times B, (a_1, b_1)(a_2, b_2) = (a_2, b_2)(a_1, b_1)$
 $\Leftrightarrow \forall a_1, a_2 \in A, b_1, b_2 \in B, a_1 a_2 = a_2 a_1$ and $b_1 b_2 = b_2 b_1$ $\begin{matrix} \parallel \\ (a_2, b_2)(a_1, b_1) \\ \parallel \\ (a_2 a_1, b_2 b_1) \end{matrix}$
 $\Leftrightarrow A, B$ are both Abelian.

1.1.31 Suppose that $T = \{x \in G \mid x \neq x^{-1}\}$, thus $\forall x \in T, x^{-1} \in T$
 therefore the finite set T can be separated into several pairs in form of
 $(a, a^{-1}) \Rightarrow |G - T|$ is even.
 $G - T$ has more than 1 element, thus there exists $a \neq e \in T$,
 hence that $a = a^{-1}$, thus $a^2 = a a^{-1} = e \Rightarrow a$ has order 2. ✓

1.1.33 (BONUS) ① Suppose that $x^i = e, x^{-i} = x^{n-i}$, WLOG $i \leq \frac{n-1}{2}$, thus $x^{n-2i} = x^{n-i} \cdot (x^i)^{-1} = e$
 Contradicts with the minimality of n .
 ② if $i \nmid k, x^i = x^k = (x^k)(x^i)^{-1} = x^{-k} = x^{-i}$.
 if $x^i = x^{-i} \Rightarrow x^{(n-2i)} = e$. the minimality of n implies $i = \frac{n}{2} = k$

1.2.3 $\forall x \in D_{2n}$ that is not a power of r , denote that $x = sr^a$.

thus $x^2 = sr^a sr^a = s(sr^{-a})r^a = s^2 = 1$.

$\langle r, sr \rangle$

that is, the order of x is 2 because $x \neq 1$.

1.2.16. replace x_i by r , y_i by s we get $r^2 = s^2 = 1, r^2 s^2 = 1$

~~what's more, $rs = (rs)^{-1} = s^{-1}r^{-1} = sr^{-1}$~~ what's more, $(rs)^2 = 1 \Rightarrow rs = (rs)^{-1} = s^{-1}r^{-1} = sr^{-1}$
 $rs = sr^{-1} \Rightarrow rsrs = sr^{-1}rs = s^2 = 1 \Rightarrow (rs)^2 = 1$

therefore, $r^2 s^2 = 1$ is equivalent to $rs = sr^{-1}$

therefore, $\{x_i, y_i \mid x_i^2 = y_i^2 = 1, (x_i y_i)^2 = 1\}$ is exactly D_4 .